

Welcome All to the  **AEA**



ARTS 7000

Training

Agenda

1. Introduction to the **ARTS-7000**
2. Walk-thru menu & screens
3. Demonstration of Key Tests
 1. ILS, VOR,
 2. Transponder + Pulse Alignment Tool
 3. Altitude monitoring
 4. CPDLC
4. Tips & Tricks
5. Q/A

ARTS 7000



Capabilities

ARTS 7000

NAV COM	PULSE	GPS VSWR 429	RAD ALT
ILS VOR Marker VHF/HF SelCal ELT CPDLC / VDL M2	DME ADS-B In/Out Mode S / ATCRBS TCAS / TIS UAT / FIS-B / TIS-B Altitude Monitoring	GPS Simulator VSWR Meter Distance to Fault ARINC 429 Transceiver*	Radio Altimeter*

SYSTEM & ACCESSORIES

Wireless Controller, Removable Battery, Power Meter,
Frequency Scanner, Antennas, Couplers, Carrying Case

ARTS 7000: Modular Hardware



	Description
1	Removable battery
2	Power Switches and Connections
3	Nav/Com Module – COM, MKR, ELT, VOR
4	Pulse Module – Transponder, DME, TCAS
5	GPS & VSWR Module
6	Expansion Slot – RadAlt

Accessories



1. Accessory Storage case



2. Antenna Mounting Tripod



3. Battery

- Telescope antenna.
- Pulse antenna.
- Coax cable (20 ft typical).
- Gorilla pod.
- Mounting plate for telescoping antenna.
- Battery pack.
- Transponder coupler. (optional)
- GPS coupler (optional)



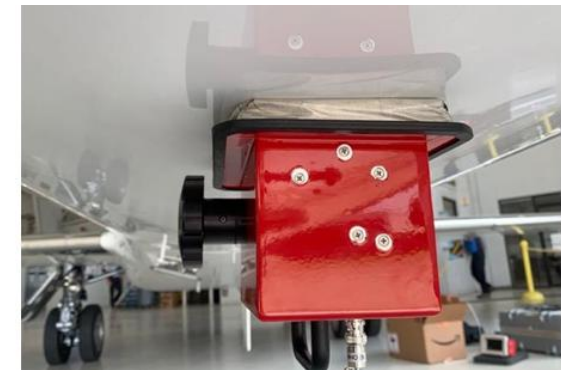
1. NAV/COM Telescopic Antenna in operation on an ARTS 7000 Test Set



2. Telescopic Antenna



3. NAV/Com Telescopic Antenna tripod mounting adapter



ARTS-7000 Application Walk-thru

ARTS 7000 User Interface

Function Menu

Back

Functions Menu

NAV

COM

ELT

DME

ATC

TCAS

Altitude Monitoring

GPS

RF Tools

Nav Sub Menu

Back

NAV Menu

ILS

MKR

ILS & MKR

VOR

Com Sub Menu

Back

COM Menu

AM

FM

SSB/HF

SelCal

ARTS 7000 User Interface

ILS

Exit

ILS

LOC Freq.110.100 MHz

GS Freq.334.400 MHz

LOC Power13.00 dBm

GS Power0.00 dBm

Ident

Off

Tone

Code

LOC Phase0 Deg

GS Phase0 Deg

LOC Tone DeleteBoth On

GS Tone DeleteBoth On

LOC0.000

GS0.000

Visual Representation Only

ArrowsKeypad

↑

←○→

↓

Default

Config

Port Ant

Auto

Start Test

📄

📶

VOR

Exit

VOR

Frequency112.200 MHz

Power0.00 dBm

Modulation30 %

Ident

Off

Tone

Code

TOFROM

30Hz Mod0 %

Tone DeleteBoth On

9960Hz Mod0 %

BEARING

-0.0+

Auto

Visual Representation Only

Default

Config

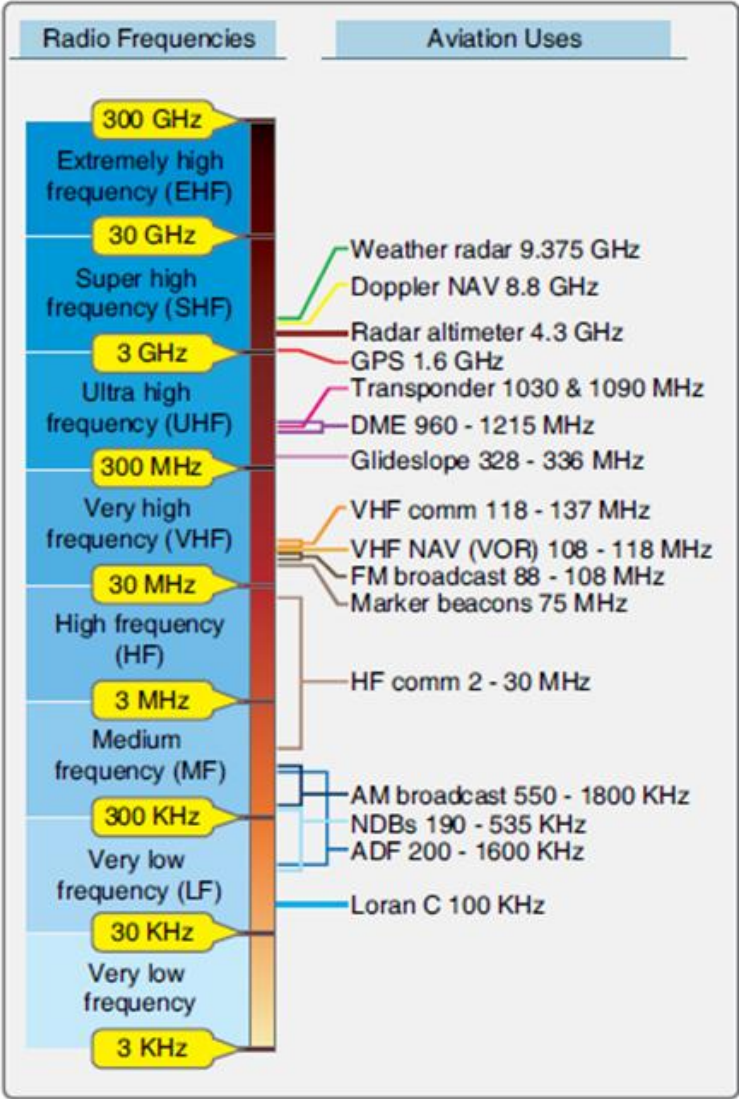
Port Ant

Start Test

📄

📶

Frequency bands where ARTS 7000 operates



COM TX

10:32 AM Mon Aug 11

Exit

AM26%

Frequency123.450 MHz

Power-14.00 dBm

Modulation30 %

Ident

Off

Tone

Code

RXTX

Frequency-

Modulation-

Power-

Audio Freq-

DefaultConfig

Stop Test

COM RX

10:32 AM Mon Aug 11

Exit

AM26%

Frequency123.450 MHz

Frequency123.449984 MHz

Power41.616 dBm

RXTX

Modulation78.46 %

Audio Freq324.670 Hz

Diagnostic Tools

Counter: 123.450000 MHz

Peak Power: 45.093 dBm



DefaultConfig

Stop Test

11:28 AM Wed Aug 6

Exit

SelCal



Frequency

123.450 MHz

Power

0.00 dBm

Modulation

85 %

Selcal Tone

AA-AA

TX Mode

Repeat

Default

Config

 Port Ant

Stop Test



Important Notes

- To operate the ELT in the Test mode, set the ELT switch to the TEST position for 3 to 10 seconds and then return it to the OFF (ARM) position. The 406 MHz data will be received within 2 seconds.
- To operate the ELT in the ON mode, set the ELT switch to the ON position for a minimum of 50 seconds and then return it to the OFF (ARM) position. The 406 MHz data will be received approximately 45 seconds after the ELT switch is set to the ON position.

WARNING

When the ELT is transmitting through its antenna and is operated in the ON mode for a period of more than 45 seconds, the 406 MHz data transmitted by the ELT will be picked up by the satellites and could trigger a Search and Rescue operation. Each region in the world has a designated time slot of 5 minutes during which the satellites will ignore any detected ELT data. Therefore, the ON-mode test using antenna must be conducted only during this designated time slot. Violating this restriction can result in severe penalties.

Please press **Start Test**

Exit

ELT Beacons

ELT 406 MHz

26%



DATA

Full Message: D6E32 49314 E53B0 F50EE 8FE0F F0F61**Hex ID:** ADC64 92629 CA761**Protocol Name:** User Location Aviation**Message Format:** Long**Protocol Flag:** User**Country Code:** 366 - United States**Aircraft Address:** N47LF**ELT Number:** 1st ELT**Radio-Locating:** 121.5 MHz**Position Data Source:** External**Latitude Encoded:** 127° 0' 0" (Default)**Longitude Encoded:** 255° 0' 0" (Default)

RESULTS

DATA: **Received**POWER: **33.59 dBm**FREQUENCY: **406.0277 MHz**Press **CLEAR** to discard the data and perform another test.

Clear

Config



Stop Test



Exit

DME

28%



Reply Freq.

1124 MHz

NAV Freq.

110.050 MHz

Channel

X

Y

Power

-2.00 dBm

Rate KTS

150 Knots

Range NM

50.0 NM

% Replies

100 %

TX Frequency

1060.92 MHz

TX Power

52.251 dBm

P1 Width

3.42 uSec

P2 Width

3.40 uSec

P1-P2 Spacing

35.85 uSec

PRF

24.84 Hz

Pause

-

IN

OUT

Default

Config



Stop Test



Location of ARTS 7000 WHEN TESTING

Usually, the unit can sit beside the aircraft. A distance of 10 ft minimum allows the best performance of the RF signals that are generated.



Back

ATC Menu



Mode S

ADS-B

ATCRBS

UAT



TIS-B

Identity
Check

92.0 %

-71.750 dBm

100 ft



Port Cable

List Test



9:27 AM Mon Aug 11

Mode S

29%

Exit

Frequency

1089.967 MHz

Top MTL

-75.500 dBm

Tail Number

N75RM

Flight ID

LAVUS505

A Code

1200

Altitude C

0 ft

Country

United States

Level

Level 2

Replies

94.0 %

Top ERP

54.565 dBm

Mode S Address

AA1933

Flight Status

Pass >

S Code

1200

Altitude S

25 ft

ADSB Available DF17

Available

Squitter Rate

1.23 sec

Config

Port Cable

GPS

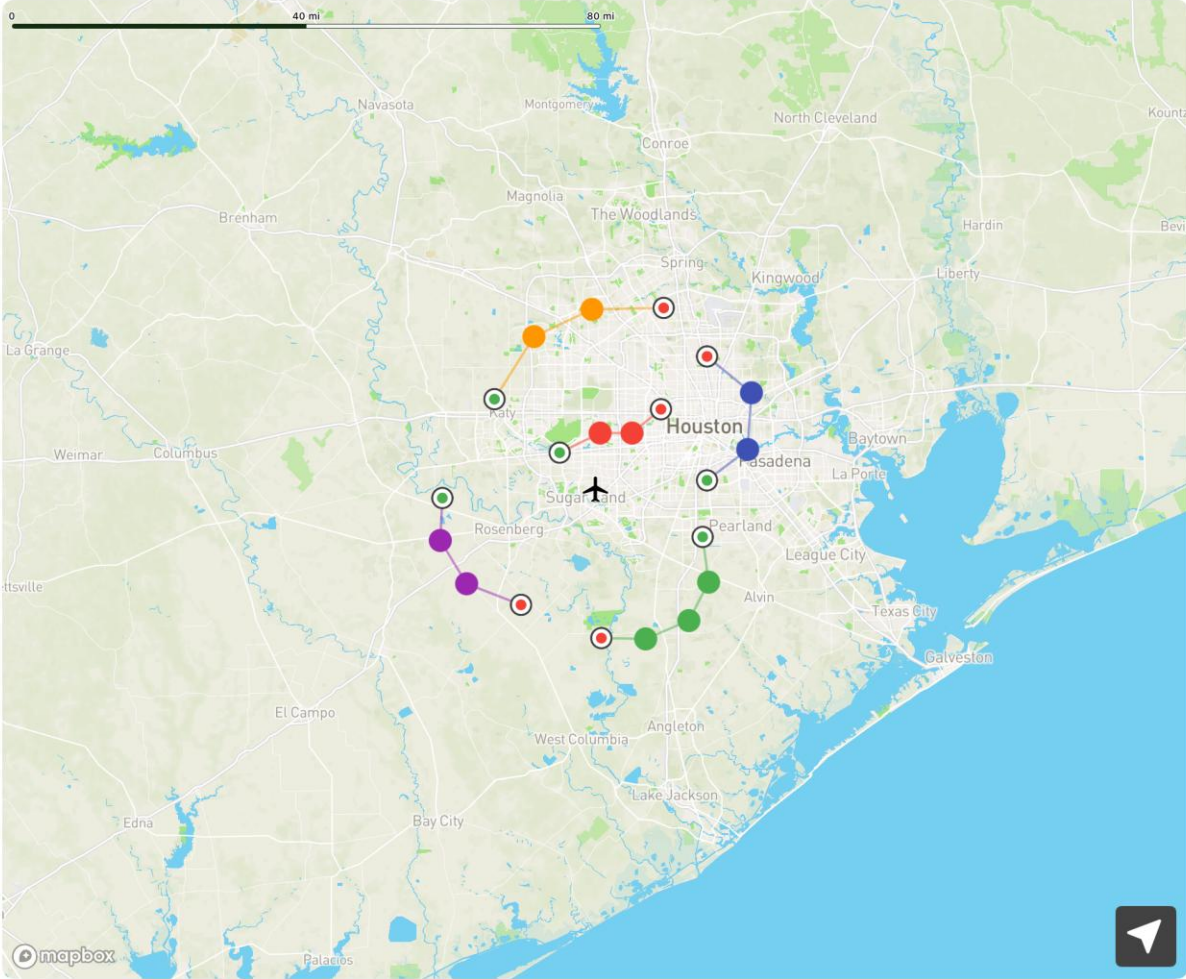
Start Test

List Test

Exit

ADS-B In





Visual representation

Config

 Port Ant

 GPS

Start Test



GENERAL

Mode

Map View

Radar

Scenario

Default

>

Power

-10.00 dBm

Save

Save as New

Begin

[Save]: Save the current scenario.

[Save as New]: Create a new scenario with the current settings.

[Pause/Resume]: Pause or resume targets after test has begun.

UUT

Latitude

29° 38' 7" N

Longitude

95° 36' 49" W

Altitude

1000 ft

[Drag & Drop]: Move UUT position on Map View.

TARGETS

T1

T2

T3

T4

T5

Enabled



Flight ID

LAVUS00X

Address

A4327X

State

Air

Ground

laversab.com



Exit

ADS-B Out

Aircraft Address		Tail Number	
Aircraft Status		Altitude (Baro)	
Aircraft Latitude		Aircraft Longitude	
Available DF17		Navigation Accuracy Category for Velocity (NACv)	
System Design Assurance (SDA)		Source Integrity Level (SIL)	
Navigation Accuracy Category for Position (NACp)		NIC	

Config

Port Ant

GPS

Start Test

List Test

Exit

Altitude Monitoring



Targets

Pitot

←

Value

→

←

Rate

→

Static Ps1

←

Value

→

←

Rate

→

Static Ps2

←

Value

→

←

Rate

→

Link

Gnd

Cancel

Go

Profile

FAR 91.411

Run

End

Next

Open

XPDR

Altitude S

S Code

Altitude C

A Code

Tail Number

Leak

Config

 Port Ant

Start Test

+





Exit

Altitude Monitoring

Targets

Pitot

Static Ps1

Static Ps2

Link

Actual

Value

Exit

Link ADTS

Profile

FAR 91.411

Open

Run

End

Next

XPDR

Altitude S

S Code

Altitude C

A Code

Tail Number

Leak

Config

Port Ant

Start Test

ADTS Setup Required

- Set Pitot units to 'Knots' and Static units to 'Feet'.
- If the ADTS has 3 outputs, set Ps2 units to 'Dfin' and the Ps2 target value to '0.0'
- Connect the ADTS to the aircraft and perform low-level leak checks.
- After low-level leak checks are successful, set the ADTS to Control mode on all outputs.
- Connect the ADTS to the ARTS using the serial cable provided, then press 'Proceed'.

Proceed

4:02 PM Mon Aug 11

Altitude Monitoring



Targets

Actual

Pitot

20.0 knots

← Value →

-19.2 knots

200.0 kts/min

← Rate →

0.4 kts/min

Static Ps1

0 ft

← Value →

63 ft

2000 ft/min

← Rate →

16 ft/min

Static Ps2

← Value →

← Rate →

Unlink

Gnd

Cancel

Go

Leak

Profile

FAR 91.411

Open

Run

End

Next

XPDR

Altitude S

100 ft

S Code

1200

Altitude C

100 ft

A Code

1200

Tail Number

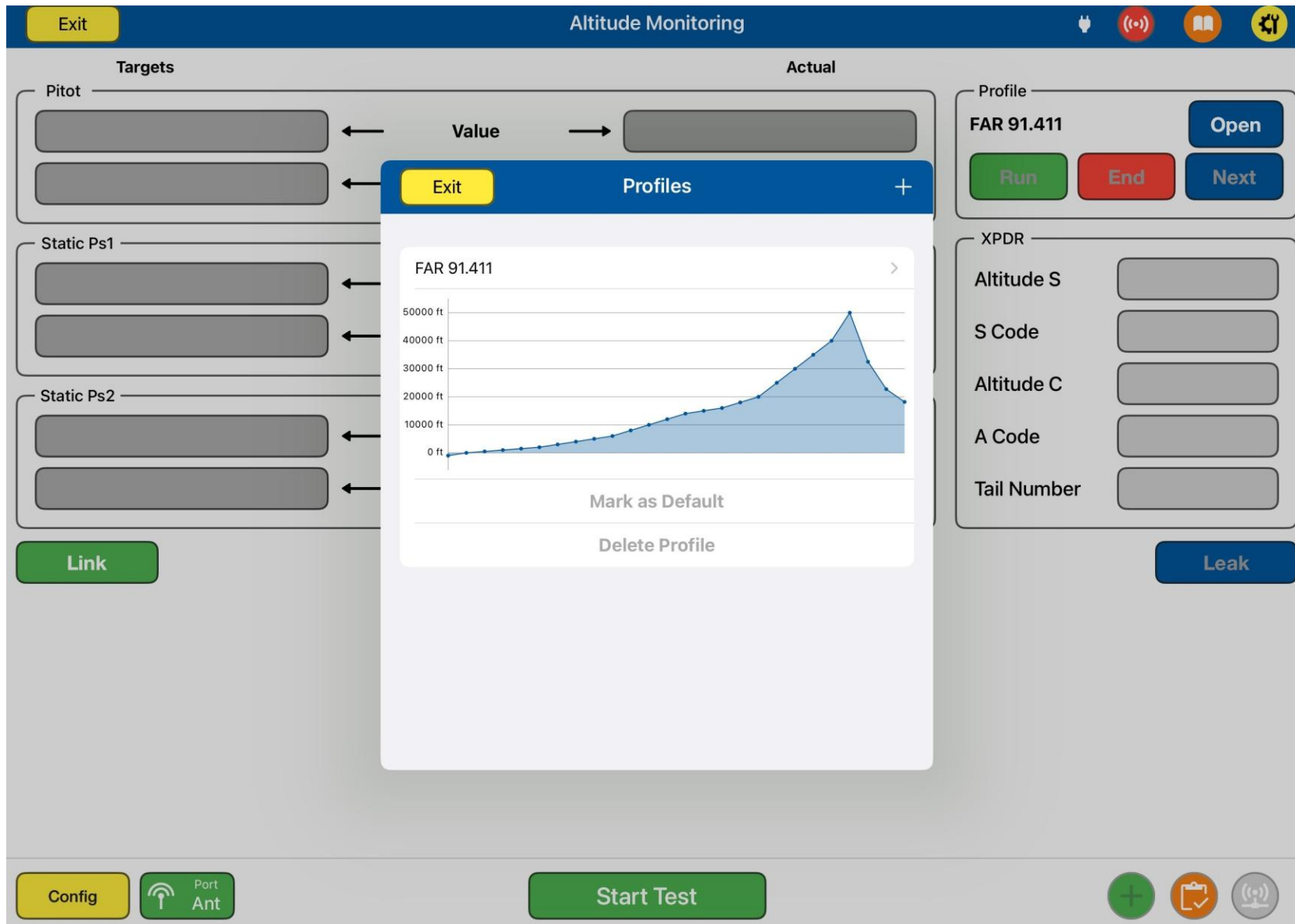
N75RM

Config

Port Cable

Stop Test





Exit

TCAS



Scenario: +2500 ft Collision

Run Scenario

Select

Save

TCAS Type

TCAS 1

TCAS 2

Intruder

Mode S

ATCRBS

MTL

Off

On

Power

-70.000 dBm

% Replies

100 %

Alt Detect

Off

On

Converge

Off

On

Range Start

15.00 NM

Range Stop

0.00 NM

Range Rate

300 kts

Alt Start

2500 ft

Alt Stop

0 ft

Alt Rate

833 ft/min

UUT ALT

10000 ft

Default

Config



Start Test

Monitor



Exit

TIS



Scenario: Default

Visualization

Select

Save

	Target 1 	Target 2 	Target 3 	Target 4 	Target 5 
Bearing	42 Deg	78 Deg	270 Deg	300 Deg	330 Deg
Range	2.0 NM	1.0 NM	3.0 NM	2.0 NM	4.0 NM
Altitude	100 ft	1000 ft	500 ft	-300 ft	-500 ft
Alt. Rate	Climbing	Descending	Level	Descending	Climbing
Heading	90 Deg	45 Deg	225 Deg	135 Deg	315 Deg
Traffic	Proximity	Traffic	Proximity	Traffic	Proximity

Read-Only

Aircraft Address

Tail Number

TSCR

TSDR

Update

UUT Heading

0 Deg

Power

-10.00 dBm

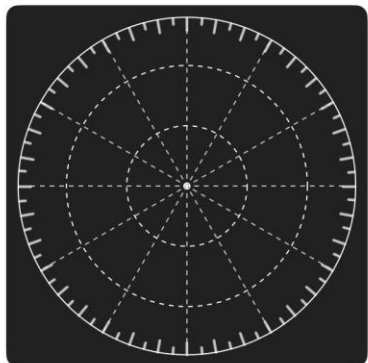
Config

 Port Ant

Start Test



Simulation



Local Date

2025-08-11

Local Time

15:32:07

Mode

Fixed

Motion

Time Zone

Local

UTC

Satellites

List

Diagnostic

Check

Power

-80 dBm

Motion Position

Routes

Houston

Speed

100 knots

Monitor

View

Config

Port
Cplr

Start Test

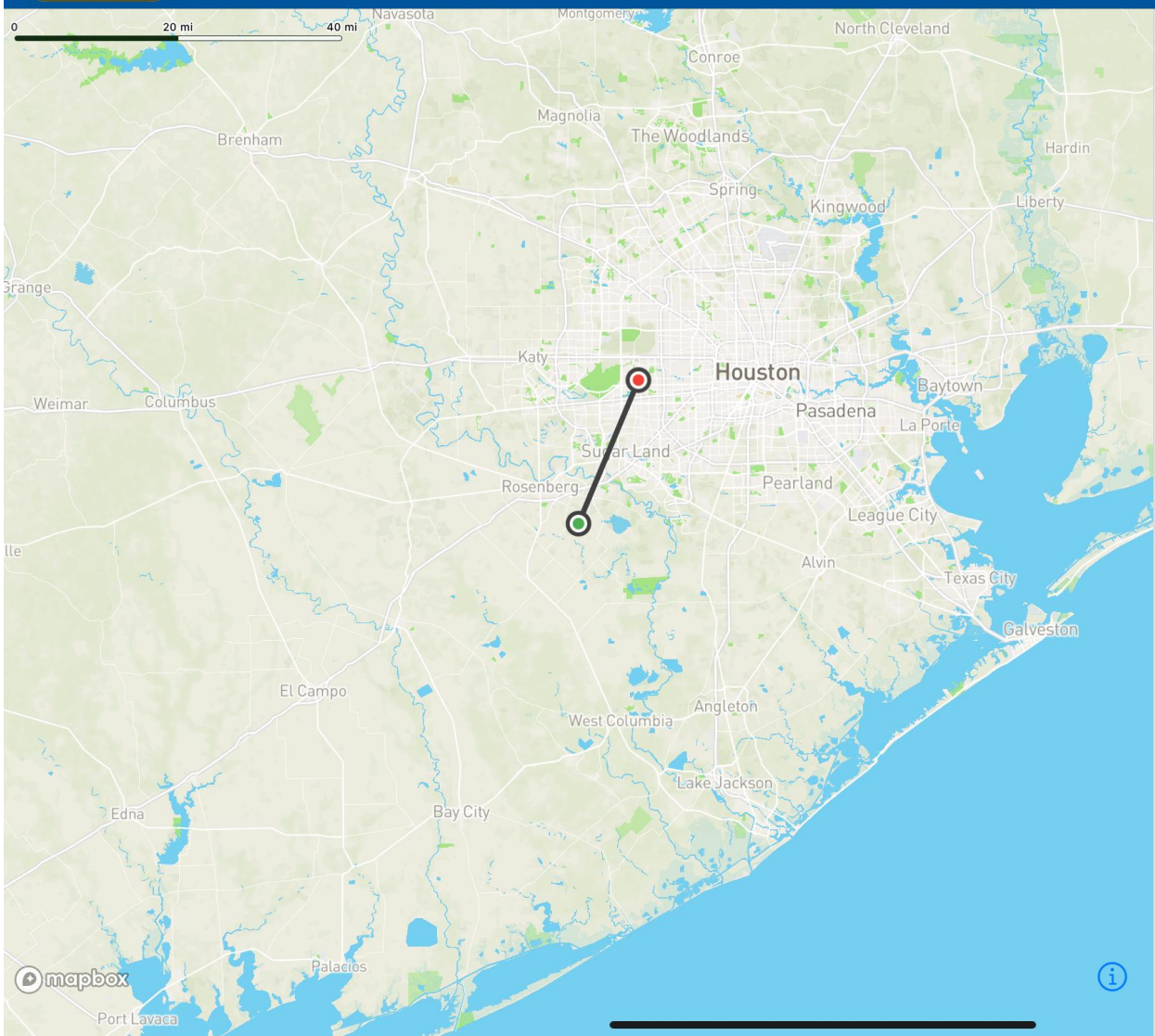


11:30 AM Wed Aug 6

100%

Exit

Monitor



STATE

Status:

Progress:0.0%

CURRENT POSITION

Latitude:-

Longitude:-

Altitude:0 ft

ACTIONS

Start

Back

Tools Menu



Counter &
Meter

DTF

VSWR / LOSS

Exit

Counter & Meter

CONFIG

Power (dBm | Watts)

dBm

RESULTS

Frequency Counter

-

Power Meter

-



Start Test

Back

Tools Menu



Counter &
Meter

DTF

VSWR / LOSS

Exit

Frequency Scanner



SETUP

Frequency Min: 120.000 MHz

Frequency Max: 125.000 MHz

Power Min: -5.000 dBm

CONFIG

Power (dBm | Watts) dBm

FREQUENCIES

There are no frequencies to show



Start Test

Exit

VSWR / LOSS



Type

VSWR

Loss S21

Frequency Range

237 MHz - 247 MHz

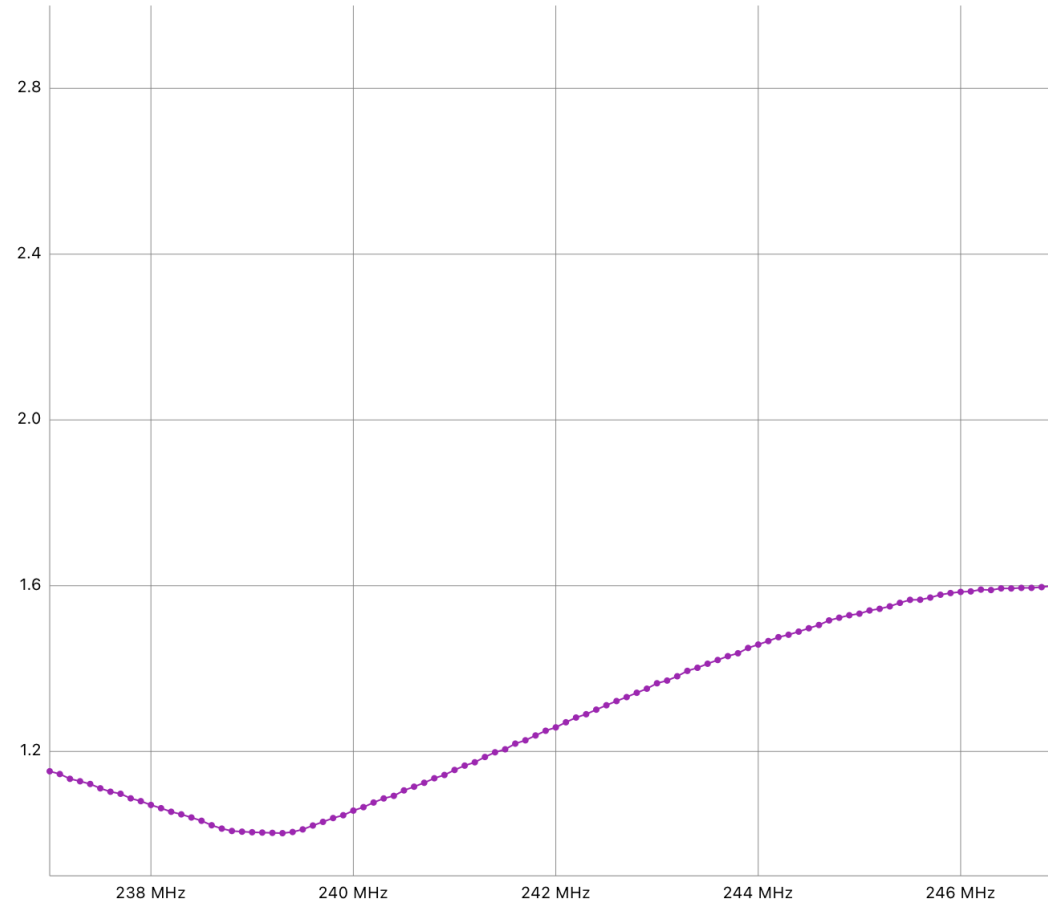
Datapoint

Frequency:

VSWR:

-

-



Stop Test



Exit

Distance To Fault

Frequency Range

1 MHz - 10 MHz

Peak (Distance): -

Datapoint

Distance: -

Exit

Frequency Range

HF Low	1 MHz - 10 MHz >
HF High	10 MHz - 30 MHz >
VHF Low	30 MHz - 50 MHz >
VHF Mid	50 MHz - 100 MHz >
Marker Beacon	72 MHz - 78 MHz >
VHF NAV	108 MHz - 118 MHz >
VHF COM 1	117 MHz - 157 MHz >
VHF COM 2	116 MHz - 126 MHz >
VHF High	155 MHz - 175 MHz >
UHF Low 1	200 MHz - 400 MHz >
UHF Low 2	237 MHz - 247 MHz >
GS	327 MHz - 337 MHz >
UHF High	400 MHz - 410 MHz >
DME	960 MHz - 1260 MHz >
XPNDR/TCAS	1030 MHz - 1090 MHz >
GPS L1	1565 MHz - 1585 MHz >

Config

Start Test

Exit

Distance To Fault



Frequency Range

1030 MHz - 1090 MHz

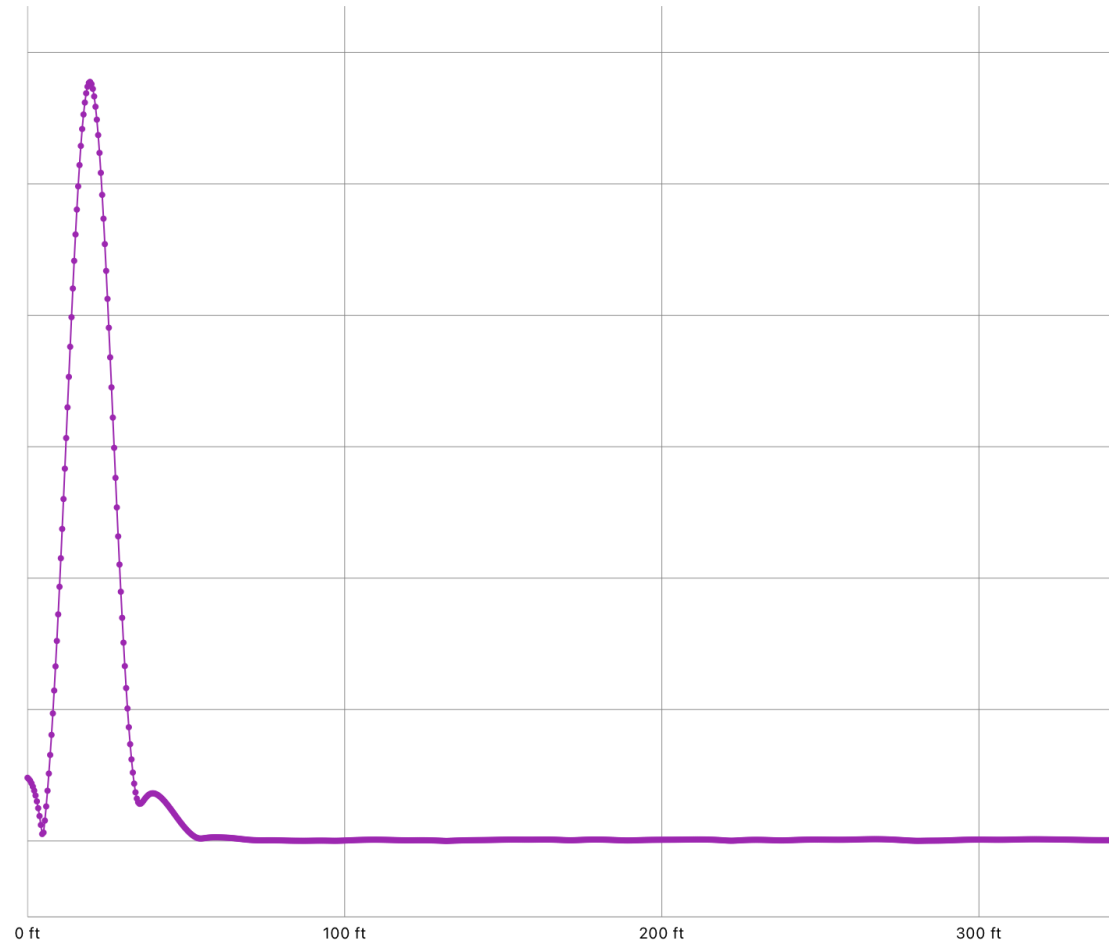
Peak (Distance):

19.7 ft

Datapoint

Distance:

-



Config

Stop Test



CPDLC

— Aircraft —
Address: A8DC84

Ground Station
Address: 100001
Code: LKSU

```

sequenceDiagram
    participant GSIF
    participant CLE
    participant RLE
    participant LOGON
    participant Connected
    GSIF --> CLE
    CLE --> RLE
    RLE --> LOGON
    LOGON --> Connected
  
```

Aircraft log on

Message data: CURRENT DATA AUTHORITY

Send



Port Cable

Stop Test

Tips & Tricks & FAQs

Tips:

1. Do not Blow up the Final! There is an Antenna Port and a Cable Port, know the difference. Antenna=Antenna, Cable=Direct Connect to the Radio or Coupler. External Fuse Coming Soon!
2. Take your time, read the prompts. They are there as a **WARNING**, do not be in too big of a hurry!
3. Pay attention to the prompts as to your connections. They will tell you how to connect and when you may need to add the external attenuator! There are even Pictures!
4. Configuration; Set it and forget about it? NO, you may have changed cables at some point, just verify the dB loss.
5. Use Profiles, they are your friends and will assist in getting through your tasks easier!
6. If in doubt push the Training Button on top or give Laversab a call, do not get frustrated, its easy!

FAQs

Question	Answer
What is the SSID or Wi-Fi for my unit?	ARTS-XXXXXX SSID always starts with the word ARTS and following the serial number of the unit.
What is the Wi-Fi password?	laversab
Insertion loss of the transponder coupler (or loss)?	Typical loss is from 0.6 to 0.9, depending on the installation can go up to 1.5 dB.
Can we connect the ELT directly to the direct port?	Yes
Can we use direct connection with the transponder?	Yes, if the transponder doesn't exceed 1KW of output power.
What serial cable works with the ADTS and ARTS 7000?	Cable Male to Male 9 Pin Straight Through.
If I delete the App, will I lose all my records?	Yes
Can we do diversity for transponder checks?	Yes
Does the GPS simulator have Beidu/Glonass/Galileo?	No, we currently support GPS L1 only.
Does ARTS 7000 communicate with the legacy pitot static unit?	Yes, certain Rev. levels do, but the support & functionality is limited.

FAQs (continued)

Question	Answer
Should I use the 10 dB attenuator with the transponder coupler?	It is for protection only when the user doesn't know the max output power. If the transponder doesn't exceed 1KW output the test can be done without the attenuator.
Should I update firmware to activate TCAS or UAT?	No, the activation of these two options have been always available. Please follow the procedure of activation that is posted on: ARTS-7000-TCAS-UAT-Activation-Procedure.pdf (laversab.com)
How often does the App gets updated?	Usually there is an update every month. We are continuously trying to improve the experience of the user and at the same time the development team works of clearing the bugs reported by customers.
Do I need to update my pitot static unit or the ARTS to make them talk to each other?	No, since release, integration to all ADTS's should be ready to communicate.
Do I need a coupler to run my transponder checks?	It is not required unless it is needed by your local authority. In terms of functionality is irrelevant for the system.
My powers are low when running a transponder check, how can I fix that?	Please check loss of the cables, antenna gain and distance (if pulse antenna is used) are set correctly. If the settings are correct a low power measurement could mean a bad transponder/antenna or installation.

Troubleshooting

Issue	Solution
My ARTS 7000 doesn't communicate with the iPad:	Check the privacy setting of the iPad and allow the local network services to the ARTS 7000 app.
Antenna alignment shows 0% replies:	Check that ARTS 7000 has the INTR (amber led on). Verify that the transponder is ON and ready to be interrogated, in some cases it needs WOW (weight on wheels) to start working. When working RPLY led shows the replies from the transponder (white).
My ARTS 7000 doesn't detect one or more modules installed:	The user can try to close the application and wait for 30 seconds to retry the connection. If this doesn't work, please contact Laversab support.
ARTS 7000 doesn't get the 406 MHz stress signal of the ELT:	When starting the test please allow ARTS to initialize the ELT mode and after it is ready please use the TEST BUTTON/ON in the ELT to trigger the signal. If it doesn't work, try one more time. If no response, please contact Laversab support.
My ARTS 7000 seems to fail or not to work in a specific test:	At any time, the user can share data of the test and send it to the Laversab cloud. Please use the GEAR icon located on the TOP RIGHT area of the app. After the data is reviewed, we will contact you to evaluate possible solutions or give a diagnostic.

Q & A

OEM	Laversab	VIAVI	TIC	CCX
Model	ARTS-7000	AVX-10K	SDR-OMNI	T-RX RP+
VOR	X	X	X	X
LOCALIZER	X	X	X	X
GLIDESLOPE	X	X	X	X
VHF COM	X	X	X	X
MODE A/C XPDR	X	X	X	X
MODE S XPDR	X	X	X	X
ADS-B OUT	X	X	X	X
ADS-B IN	X	X	X	X
ADS-B UAT	X	X	X	X
DME	X	X	X	X
TACAN	X	X	Military Opt.	
ELT	X	X	X	X
HF COM	X	X	X	X
SEL CAL	X	X	X	X
ARINC 429 DB	Coming Soon			X
TCAS	X	X	X	X
VDL MODE 2	Coming Soon			
VSWR	X	X	X	X
DISTANCE TO FAULT	X	X	X	X
EXPORT TEST REPORTS	X	X		X
GPS	X	Coupler Based	Coming Later \$7.5	Option
RADAR ALTIMETER	Coming Soon			
MILITARY IFF 2/3/4/5	In Work		No 4/5 Military	
BASE PACKAGE PRICE				

Some of the Great OEM's we work with:



Get in Touch

Email	Sales@laversab.com
Phone	281-325-8300
Address	Laversab Inc (HQ) 505 Gillingham Lane Sugar Land, TX, 77478 USA